

AI-Based Automotive Review and Customer Sentiment Analytics

AI/ML Technologies: BERT, XLM-R, NLP | Comprehensive Project Report

1. Project Goals & Problem Statement

Modern automotive customer reviews express multi-faceted opinions across distinct vehicle systems. A single review frequently contains contrasting sentiments for different vehicle components: *'The battery range is excellent, but the charging time is too long.'* Standard document-level sentiment classification assigns a single unified score, obliterating the crucial engineering insight that battery range is performing well while charging infrastructure requires optimization. This project implements an end-to-end NLP and Transformer pipeline to extract individual vehicle aspects and infer fine-grained aspect-level sentiment.

2. System Architecture

The project follows a modular 7-stage NLP architecture:

Stage	Pipeline Stage	Description / Implementation
1	Automotive Reviews	Unstructured genuine customer written reviews and ratings
2	Text Preprocessing	Whitespace normalization, boundary parsing, negation preservation
3	Tokenizer	BERT WordPiece & XLM-R SentencePiece BPE tokenizers
4	BERT / XLM-R	Pretrained transformer encoders fine-tuned for automotive sequence classification
5	Sentiment Classification	Calibrated 3-class prediction: Negative (0), Neutral (1), Positive (2)
6	Aspect Extraction	Decomposition across 9 automotive domains into specific aspect terms
7	Aspect-Level Sentiment	Targeted aspect polarity mapping output (e.g. Battery range -> Positive)

3. 9 Core Automotive Review Aspects

#	Aspect Domain	Target Sub-Aspect Terms	Sample Review Context
1	Vehicle	Build quality, Exterior styling, Handling, Design	Fit and finish, road presence
2	Engine	Power, Horsepower, Torque, Acceleration, Turbo	Engine pickup and highway passing
3	Battery	Battery range, Charging time, Battery health, Wallbox	EV range vs DC fast charging speed
4	Mileage	Fuel efficiency, Gas mileage, KMPL, MPG	City vs highway fuel consumption
5	Safety	Braking system, Airbags, Driver assist (ADAS), Crash test	Active emergency braking & safety
6	Comfort	Ride quality, Cabin quietness, Seat comfort, Suspension	Rear seat legroom and cabin noise
7	Service	Dealership service, Maintenance, Customer support	After-sales support and repair turnaround
8	Infotainment	Touchscreen, Apple CarPlay, Android Auto, Sound system	Dashboard screen responsiveness
9	Price	Value for money, Purchase price, Maintenance cost	Sticker price vs ongoing operating costs

4. Canonical Specification Verification

Input Review: "The battery range is excellent, but the charging time is too long."

Aspect	Battery range	Charging time
Sentiment	Positive	Negative

Result: The syntactic clause segmenter cleanly split the contrastive conjunction 'but', associating 'battery range' with the positive evaluation clause and 'charging time' with the negative constraint, achieving 100% adherence to specification.